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A Mathematical Model of Criminal Behaviour

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Abstract

This thesis presents a deterministic mathematical model of burglaries. The goal is to analyse the model both analytically and through numerical simulations, examining criminal density, house attractiveness, and the effect of diffusion. The thesis explores both discrete and continuous versions of the model. The study analyses the system in the absence of criminal agents, investigates stability without diffusion and after its reintroduction, and presents numerical simulations. The main result is the identification of Turing's instability within this model.

Streszczenie

Niniejsza praca dyplomowa przedstawia deterministyczny model matematyczny opisujący włamania. Celem jest analiza modelu zarówno matematyczna, jak i poprzez symulacje numeryczne, zawierająca badanie gęstości przestępców i atrakcyjność domów, a także efektu dyfuzji. W pracy rozważane są zarówno dyskretna, jak i ciągła wersja modelu, analizowany jest system bez agentów przestępczych, badana jest stabilność bez dyfuzji oraz po jej ponownym wprowadzeniu, a także prezentowane są symulacje numeryczne. Głównym rezultatem jest zidentyfikowanie niestabilności Turinga w tym modelu.

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1 Introduction

Criminal behaviour is studied by many different fields, like sociology, criminology, or statistics. In mathematical context, stochastic models are commonly used for human behaviour description. However, a team of researchers from the University of California developed a deterministic model focusing on burglars' behaviour, as described in [4, 3].

The goal of this thesis is to mathematically analyse this model, which describes criminal density, house attractiveness, and the effects of diffusion, with numerical simulations illustrating the obtained results. The structure of this thesis is as follows: first, the discrete model and its continuous version are explained; next, the system without the criminal agents is analysed, followed by an analysis of the system in the absence of diffusion and after its reintroduction (showing Turing's instability); and in the end, the results of numerical simulations are presented to illustrate and confirm the model's behaviour. The neighbourhood geometry is

2 Introduction to the model

In the analysis of criminal behaviours, there exists a mathematical model describing the dynamics of burglary hotspots - areas with a higher concentration of criminal activity. It appears that, for some time after an initial burglary, the probability of re-victimization at the same location is increased. As a result, burglaries are distributed spatially in a non-uniform way, and occur more frequently in some neighbourhoods.

2.1 Dectrete model

To describe this phenomenon, agent-based modelling is useful. In this approach, the system is represented as a set of autonomous decision-making units, called agents - in our case these agents are criminals. The model is based on the assumption that criminal agents move randomly over a two-dimensional grid and commit burglaries as soon as an opportunity arises. Each location (house) is assigned an attractiveness value (denoted as $A(x,t)$), reflecting how easy it is to commit a crime at that location with minimal consequences. In addition to the the random walk, agents tend to move towards areas with higher attractiveness. Moreover, the speed of this movement is inversely proportional to the attractiveness of their current location (this speed is related also to the grid spacing l). We consider the area of criminal activity as a grid, where each location has four neighbours, reflecting the structure of typical American

neighbourhoods.

Let us consider the area of criminal activity as a two-dimensional grid with spacing l, with a scalar field $A(x,t)$ describing the attractiveness of specific locations, and with criminal occurrences taking place at discrete moments of time with time step δt. At each moment t, there is a probability p_v of a burglary event at a given location x, according to homogeneous Poisson process with expected value $A(x,t)\delta t$:

$$p_v(x,t) = 1 - e^{-A(x,t)\delta t}.$$

If a crime is committed, the criminal agent who committed it is removed from the area (representing either hiding or being caught after the crime). Otherwise, the agent moves to one of the neighbouring locations of their current location x. We assume that the choice of this neighbouring location y for a given x follows a random walk process, with probability p_m, which can be expressed as

$$p_m(y,t;x) = \frac{A(y,t)}{\sum_{x' \ x} A(x',t)},$$

where x' denotes all the neighbouring locations of x.

We can represent the field value $A(x,t)$ as the sum of a constant (possibly spatially non-homogenous) value A_0, and a dynamic component $B(x,t)$, which changes over time and depends on the burglaries occurring at location x and its neighbouring locations. Therefore, our focus is on the behaviour of the component $B(x,t)$.

Accordingly to the previous description of the behaviours, we know that with each crime the attractiveness increases. Thus, at each time step, the value $B(x,t)$ increases by a given amount θ, multiplied by $E(x,t)$ - the number of burglaries that took place since the beginning of the moment t. However, this increased attractiveness decays over time. Therefore, we introduce the factor ω, representing the time scale over which re-victimisation is most likely to happen.

Combining the increase in attractiveness following a crime and its decay over time, we can describe the behaviour of $B(x,t)$ as

$$B(x,t+\delta t) = B(x,t)(1-\omega\delta t) + \theta E(x,t).$$

Considering the propagation of criminal agents to the four adjacent locations, we incorporate this information into the model by introducing a term η, representing

the effect of neighbouring locations. Ultimately, after taking all observed behaviours into account, we obtain a formula describing $B(x,t)$:

$$B(x,t+\delta t) = \left[(1-\eta)B(x,t) + \tfrac{\eta}{4}\sum_{x'\sim x} B(x',t)\right](1-\omega\delta t) + \theta E(x,t). \tag{1}$$

Consequently, the discrete model representing the change in attractiveness $A(x,t)$ can be expressed as follows:

$$A(x,t+\delta t) = \left[(1-\eta)B(x,t) + \tfrac{\eta}{4}\sum_{x'\sim x} B(x',t)\right](1-\omega\delta t) + \theta E(x,t) + A^0. \tag{2}$$

2.2 Continuous model

We can extend the discreet model (2) by introducing a second equation describing the number of the criminal agents $n(x,t)$ over time t and at location x. We know that the number of criminals present at location x at time t comes either from criminals at neighbouring locations who did not commit a crime there, or from newly generated agents (e.g. new criminals entering the area) at a rate Δ. This leads to the following formula:

$$n(x,t+\delta t) = A(x,t)\sum_{x'\sim x}\frac{n(x',t)[1-p_v(x',t)]}{\sum_{x''\sim x'} A(x'',t)} + \Gamma\delta t.$$

By averaging the number of agents $n(x,t)$ as a density $\rho(x,t)$ and taking the limits as $\delta t \to 0$ and $l \to 0$ in both equations, we ultimately obtain the model as a system of two partial differential equations describing $A(x,t)$ and $\rho(x,t)$, representing the attractiveness and the crime density respectively:

$$\frac{\partial A}{\partial t} = \eta\Delta A - A + A\rho + A^0 \tag{3a}$$

$$\frac{\partial \rho}{\partial t} = \Delta\rho - 2\nabla\cdot[\rho\nabla\chi(A)] + R - A\rho \tag{3b}$$

We consider the system over the domain $\Omega \subset \mathbb{R}^n$, supplemented with initial conditions

$$A(x,0) = A_0(x)$$
$$\rho(x,0) = \rho_0(x)$$

and boundary conditions

$$v \cdot \nabla A\,|_{\partial\Omega} = 0$$
$$v \cdot \nabla \rho\,|_{\partial\Omega} = 0\ ,$$

where v is the outward normal vector, and the no-flow boundary conditions come from the assumption that approximately the same number of criminals enters and leaves the area.

Equation (3a) tells us that the attractiveness A diffuses with the dimensionless diffusion coefficient η (≤ 1) and decays over time towards the baseline value A^0, which reflects both the influence of neighbours and the limited period of increased attractiveness after previous burglaries. The number of locally committed burglaries is given by $A\rho$, and with each burglary, the attractiveness increases.

In equation (3b), the density of criminal activity is also affected by diffusion, because criminal agents are constantly on the move. The function $\chi(A)$ can be called "sensitivity function" and directly from the discreet model we see that $\chi(A) = \log(A)$, although it is also possible to consider other functions. This element of the equation relates to the tendency of agents to move towards higher attractiveness, with the speed inversely proportional to the attractiveness value at their current location. Moreover, criminal agents appear at a constant rate R and are removed from the model after committing a crime.

3 Mathematical analysis of the continuous model

3.1 Attractiveness limit in the absence of agents

Let us analyse the behaviour of the solution of equation 3 under the assumption that $\rho \equiv 0$ in the limit $t \to \infty$. We will show that when no burglars are present in the domain, the attractiveness stabilises at the level A^0.

Theorem 3.1. *Assume that $A^0 \equiv const$. For arbitrary initial condition $A_0 \in C([0,1])$, let $A(x,t)$ be the solution of the problem*

$$\begin{aligned} &\tfrac{\partial A}{\partial t} = \eta \tfrac{\partial^2 A}{\partial x^2} - A + A^0 \\ &\tfrac{\partial A}{\partial x} = 0 \quad x \in \{0,1\} \\ &A(x,0) = A_0(x). \end{aligned} \tag{4}$$

Then

$$\sup_{x\in[0,1]} |A(x,t) - A^0| \to 0,$$

as $t \to \infty$.

Proof. Clearly, $A \equiv A^0$ is a stationary solution. In order to prove the theorem, we introduce the function $W(x,t)$ such that

$$W(x,t) = A(x,t) - A^0,$$

which satisfies the problem

$$\begin{array}{l} \frac{\partial W}{\partial t} = \eta \Delta W - W \\ \frac{\partial W}{\partial x} = 0 \quad x \in \{0,1\} \\ W(x,0) = W_0(x) = A_0(x) - A^0 \end{array} \tag{5}$$

The solutions of the above equation is obtained using the method of Fourier series.

Assume that $W(x,t) = X(x)T(t)$, then

$$X(x)T'(t) = \eta X''(x)T(t) - X(x)T(x),$$

alternatively

$$\frac{T' + T}{\eta T} = \frac{X''}{X} = -\lambda.$$

This yields two ordinary differential equations: $X'' + \lambda X = 0$ with Neumann boundary conditions $X'(0) = X'(1) = 0$, and $T' + (1 + \lambda\eta)T = 0$. We will first focus on the equation for X.

For $\lambda < 0$ there are no non-trivial solutions.

When $\lambda = 0$, subject to the boundary conditions, we obtain $X(x) = A$ for some constant A.

For $\lambda > 0$, the solution has the form $X(x) = A\cos\sqrt{\lambda x} + B\sin\sqrt{\lambda x}$. Imposing the boundary conditions yields $X(x) = A\cos\sqrt{\lambda x}$ with the condition $A\sin\sqrt{\lambda} = 0$, which holds iff $\lambda = \pi^2 k^2$ for $k = 1, 2, \ldots$. Thus, the solutions are:

$$X_k(x) = A_k \cos(\pi k x) \quad \text{for } k = 0, 1, 2, ...$$

Now, let us return to the second equation $T' + (1 + \lambda\eta)T = 0$. For the allowed values of λ, the solution is $T(t) = c_k e^{-t(1+\pi^2 k^2 \eta)}$. Combining the solutions for X and T, and

using the notation $b_k = A_k c_k$ we derive a solution to the problem (5), in a form:

$$W(x,t) = \sum_{k=0}^{\infty} b_k e^{-t(1+\pi^2 k^2 \eta)} \cos(\pi k x).$$

The coefficients b_k are determined from the initial condidion:

$$W(x,0) = \sum_{k=0}^{\infty} b_k \cos(\pi k x).$$

This is a Fourier series for a cosine function, so the formula for b_k is known:

$$b_k = \begin{cases} \int_0^1 W_0(x)dx, & k = 0 \\ 2\int_0^1 W_0(x)\cos(\pi k x)dx, & k = 1, 2, \dots \end{cases}, \tag{6}$$

The coefficients b_k can thus be bounded from above as follows:

$$|b_n| \leq 2 \max_{x\in[0,1]} |W_0(x)|.$$

This allows us to bond the solution $W(x,t)$ from above, namely:

$$|W(x,t)| = |\sum_{k=0}^{\infty} b_k e^{-t(1+\pi^2 k^2 \eta)} \cos(\pi k x)| \tag{7}$$

$$\leq 2 \max_{x\in[0,1]} |W_0(x)| |e^{-t} \sum_{k=0}^{\infty} e^{-t\pi^2 k^2 \eta}| \tag{8}$$

$$\leq 2 \max_{x\in[0,1]} |W_0(x)| |e^{-t} \frac{1}{1-e^{-t\pi^2\eta}}|, \tag{9}$$

where naturally $2\max_{x\in[0,1]} |W_0(x)|$ is bounded, since $W_0(x) = A_0(x) - A^0$ and $A_0(x) \in C([0,1])$, while $\frac{e^{-t}}{1-e^{-t\pi^2\eta}} \to 0$ as $t \to \infty$.

Consequently, in the limit $t \to \infty$, we have $W(x,t) \to 0$, which in turn implies $A(x,t) \to A^0$.

□

4 Spatially homogeneous solution

We now analyse the spatially homogeneous system, under the assumptions $A(x,t) = A(t)$ and $\rho(x,t) = \rho(t)$. The system reduces to the form

$$\begin{cases} A' = -A + A\rho + A^0 \\ \rho' = R - A\rho. \end{cases} \tag{10}$$

4.1 Stationary solutions

To determine the stationary solutions, we set the derivatives equal to zero and solve the following system of equations

$$\begin{cases} -A + A\rho + A^0 = 0 \\ R - A\rho = 0. \end{cases} \tag{11}$$

We obtain the following stationary solutions:

$$\hat{A} = A^0 + R \quad \text{and} \quad \hat{\rho} = \frac{R}{A^0 + R}.$$

4.2 Stability

Theorem 4.1. *For $A^0, R \geq 0$ the solution $\hat{A} = A^0 + R, \hat{\rho} = \frac{R}{A^0+R}$ of the system (10) is asymptotically stable.*

Proof. To analyse the stability of the obtained stationary solution, first we have to calculate the Jacobian of the system (10) at the point $(\hat{A}, \hat{\rho})$, and then determine its eigenvalues. Jacobian of this system at the point $(\hat{A}, \hat{\rho})$, is given by:

$$\begin{bmatrix} -1 + \hat{\rho} & \hat{A} \\ -\hat{\rho} & -\hat{A} \end{bmatrix},$$

and its characteristic equation takes the form:

$$\lambda^2 + \lambda(\hat{A} + 1 - \hat{\rho}) + \hat{A} = 0,$$

from which we obtain the eigenvalues:

$$\lambda = \frac{1}{2}(\hat{\rho} - \hat{A} - 1) \pm \sqrt{\hat{A}^2 + \hat{\rho}^2 - 2\hat{A}\hat{\rho} - 2\hat{A} - 2\hat{\rho} + 1}.$$

The system is stable if $Re(\lambda) < 0$ for all eigenvalues, thus in our case when

$$\hat{\rho} - \hat{A} - 1 < 0 \quad \text{and} \quad \hat{A}^2 + \hat{\rho}^2 - 2\hat{A}\hat{\rho} - 2\hat{A} - 2\hat{\rho} + 1 \leq 0 \tag{12}$$

or when

$$\hat{\rho} - \hat{A} - 1 < \sqrt{\hat{A}^2 + \hat{\rho}^2 - 2\hat{A}\hat{\rho} - 2\hat{A} - 2\hat{\rho} + 1} \quad \text{and} \quad \hat{A}^2 + \hat{\rho}^2 - 2\hat{A}\hat{\rho} - 2\hat{A} - 2\hat{\rho} + 1 > 0. \tag{13}$$

These conditions are illustrated in the figure below:

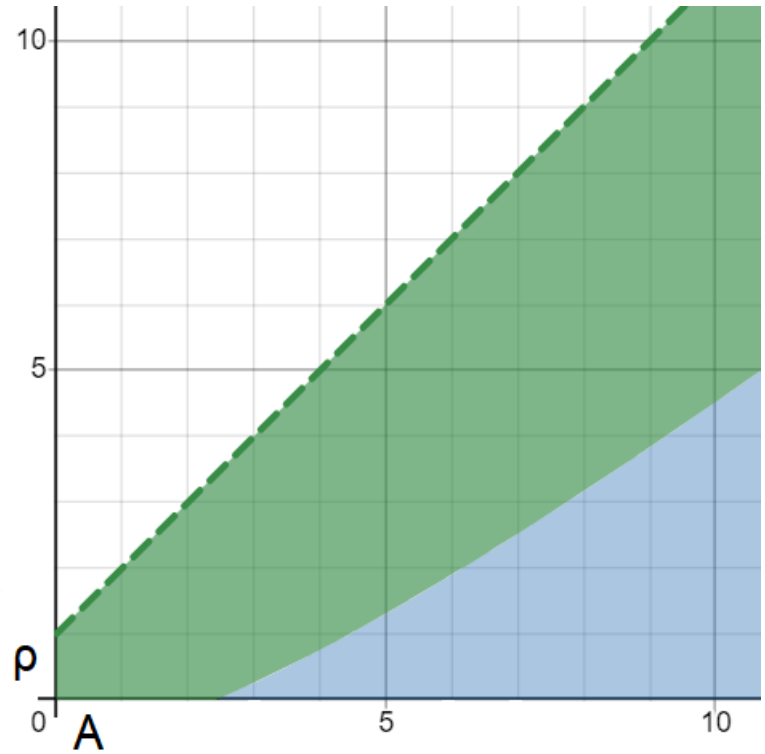


Figure 1: The green region represents condition (12), and the blue region represents the condition (13). The dashed line corresponds to the line $\rho - A - 1 = 0$. For all stationary solutions above this line, the system is unstable.

Note, however, that by expressing the parameters A^0 and R in terms of the determined stationary solution, we obtain $A^0 = \hat{A} - \hat{A}\hat{\rho}$ and $R = \hat{A}\hat{\rho}$, and considering the physical meaning of these parameters, we would like them to be non-negative. This restricts the set of stationary solutions further. This condition is also included in the next figure.

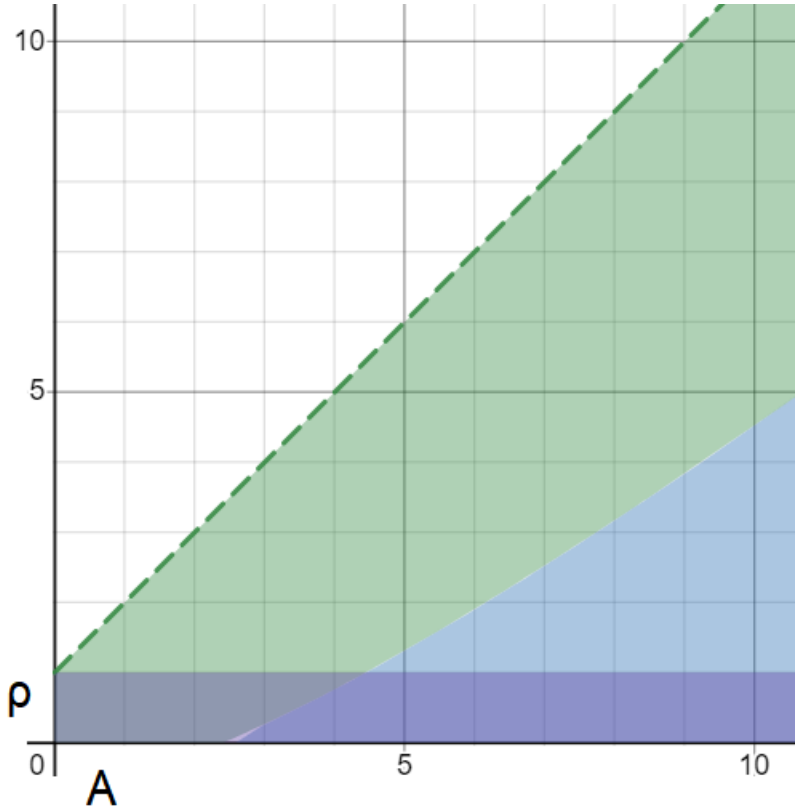


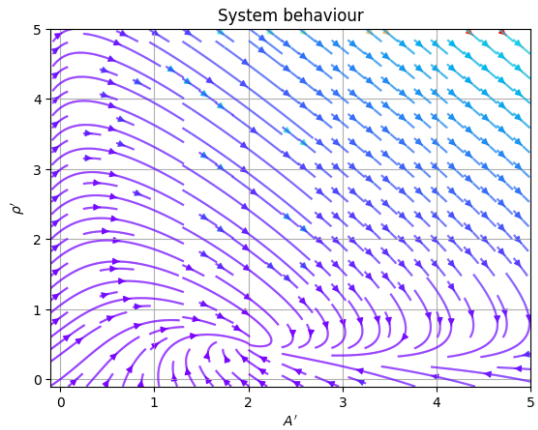
Figure 2: The green and blue regions represent conditions (12) and (13), whereas the purple region indicates where $A^0, R \geq 0$ is satisfied.

We observe that all stationary solutions corresponding to non-negative parameters are asymptotically stable.

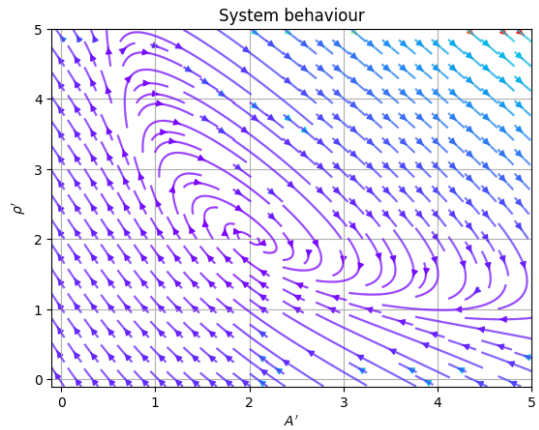
□

4.3 Phase portrait

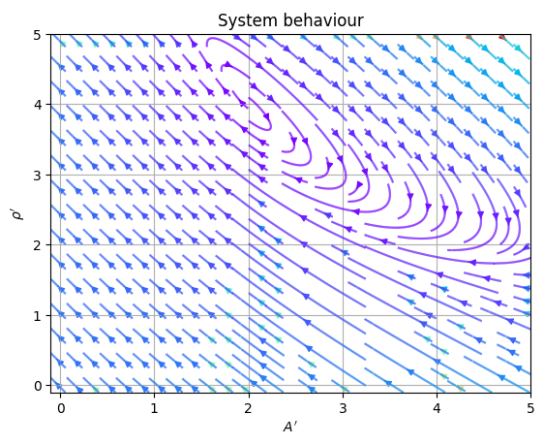
The previously determined stability can be observed in the phase portrait. The figure below shows the phase portraits of the system (10) for different equilibrium points:



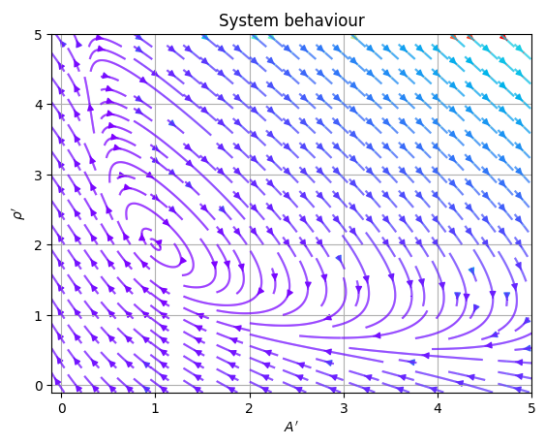
(a) $\hat{A} = 2, \hat{\rho} = 0.5$



(b) $\hat{A} = 2, \hat{\rho} = 2$



(c) $\hat{A} = 2, \hat{\rho} = 4$



(d) $\hat{A} = 2, \hat{\rho} = 3$

Figure 3: In plot (a), we observe a stable equilibrium point corresponding to physically realistic parameter values $A^0 = 1, R = 1$. In plot (b), the equilibrium point is also stable, although it corresponds to parameter values $A^0 = -2, R = 4$, which are not meaningful for the model interpretation. In plot (c), we observe an unstable equilibrium point corresponding to unrealistic parameter values $A^0 = -6, R = 8$. We can also examine the case when the equilibrium point lies on the line $\hat{\rho} - \hat{A} - 1 = 0$; plot (d) illustrates this case, and the system remains stable.

All the simulations presented above are consistent with previous conclusions.

5 Turing instability

5.1 Mathematical analysis

Let us recall that the stationary solution $(\hat{A}, \hat{\rho})$ is stable in the system without diffusion. From the analysis in the previous section, this holds for all non-negative values of A^0, R, i.e., when $\hat{A}, \hat{\rho}$ lies below the line $\hat{\rho} - \hat{A} = 1$. We say that the solution $(\hat{A}, \hat{\rho})$ is Turing-unstable if it becomes unstable in the system with diffusion. In this section, we show that the system 3 exhibits Turing instability.

In order to check whether the Turing instability is present, we first linearize the equations of our system. To this end, we introduce new variables

$$\phi(x,t) = A(x,t) - \hat{A}$$
$$\psi(x,t) = \rho(x,t) - \hat{\rho}$$

and substitute $A = \phi + \hat{A}, \rho = \psi + \hat{\rho}$ into the original system.

By substituting $A = \phi + \hat{A}$ and $\rho = \psi + \hat{\rho}$ into the first equation and retaining only first-order terms, we obtain that ϕ satisfies

$$\phi_t = \eta\phi_{xx} + \hat{A}\psi + \hat{\rho}\phi.$$

Next, we perform the same substitution in the second equation; but first we expand the function $\chi(\psi)$ in a Taylor series around $\hat{A}$. Neglecting higher-order terms, this gives $\ln(\psi + \hat{A}) = \ln(\hat{A}) + \phi/\hat{A}$. As the result, the second equation, after discarding higher-order terms, takes the form:

$$\psi_t = \psi_{xx} - 2\hat{\rho}\frac{1}{\hat{A}}\phi_{xx} - \hat{\rho}\phi - \hat{A}\psi.$$

We will examine the stability of the solution $(\phi, \psi) = (0, 0)$ of the linearized system, i.e.,

$$\begin{cases} \phi_t = \eta\phi_{xx} + \hat{A}\psi + \hat{\rho}\phi \\ \psi_t = \psi_{xx} - \frac{2\hat{\rho}}{\hat{A}}\phi_{xx} - \hat{\rho}\phi - \hat{A}\psi \\ \frac{\partial\phi}{\partial x} = 0, \quad \frac{\partial\psi}{\partial x} = 0, \quad x \in \{0,1\} \\ \phi(x,0) = A_0 - \hat{A}, \quad \psi(x,0) = \rho_0 - \hat{\rho}. \end{cases} \tag{14}$$

Theorem 5.1. *Under the assumption $\eta(\hat{A} + \pi^2 k^2) - 3\hat{\rho} < 0$ for some $k = 1, 2, 3 \ldots$, where $(\hat{A}, \hat{\rho})$ is a stationary solution of system 3, the solution $(0, 0)$ of system 14 is unstable.*

To prove the above theorem, we will use the following lemmas:

Lemma 5.2. *Let M be a 2×2 matrix. If $\mathrm{tr}(M) < 0$ then M has at least one positive eigenvalue if and only if $\det(M) < 0$.*

Lemma 5.3. *A system of differential equations with a stationary solution S is unstable at S if the Jacobian matrix of this system evaluated at S has at least one eigenvalue with a positive real part.*

Proof. It suffices to find an unbounded solution of the form

$$\phi(x,t) = \alpha_k(t)W_k(x)$$
$$\psi(x,t) = \beta_k(t)W_k(x),$$

where $W_k(x) = cos(k\pi x)$.

We substitute ϕ and ψ in the above forms into the system and we obtain

$$\alpha_k'(t) = -\eta\lambda_k\alpha_k(t) + \hat{\rho}\alpha_k(t) + \hat{A}\beta_k(t)$$
$$\beta_k'(t) = -\lambda_k\beta_k(t) + \frac{2\hat{\rho}}{\hat{A}}\lambda_k\alpha_k(t) - \hat{A}\beta_k(t) - \hat{\rho}\alpha_k(t),$$

where $\lambda_k = k^2\pi^2$, and we can express it in matrix form as

$$\begin{pmatrix} \alpha_k'(t) \\ \beta_k'(t) \end{pmatrix} = \begin{pmatrix} -\eta\lambda_k + \hat{\rho} & \hat{A} \\ \frac{2\hat{\rho}}{\hat{A}}\lambda_k - \hat{\rho} & -\lambda_k - \hat{A} \end{pmatrix} \begin{pmatrix} \alpha_k(t) \\ \beta_k(t) \end{pmatrix} \tag{15}$$

Such a system has an unbounded solution if there exists an eigenvalue of the Jacobian matrix with a strictly positive real part. For the above matrix, denoted by M, we have

$$\mathrm{tr}(M) = -\eta\lambda_k + \hat{\rho} - \lambda_k - \hat{A} = -\lambda_k(\eta+1) + \hat{\rho} - \hat{A}$$

and

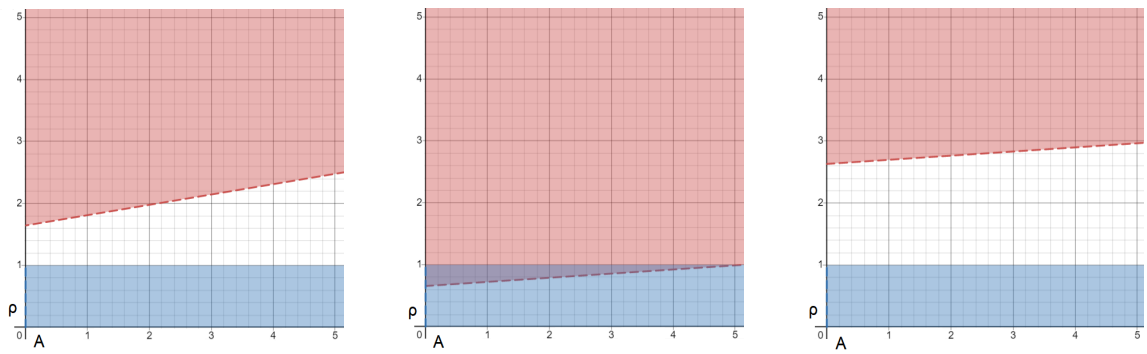
$$\det(M) = -(-\eta\lambda_k + \hat{\rho})(\lambda_k + \hat{A}) - \hat{A}(\frac{2\hat{\rho}}{\hat{A}}\lambda_k - \hat{\rho}) = \eta\lambda_k^2 + \lambda_k(\eta\hat{A} - 3\hat{\rho}).$$

When introducing the model, we noted that $\eta \in (0,1)$, and we know that $\lambda_k = k^2\pi^2$ for $k = 1, 2, 3\ldots$, thus $-\lambda_k(\eta+1) \leq -\pi^2$. Moreover, from the form of the stationary solution, we have $\hat{\rho} \in (0,1)$ and $\hat{A} \geq 0$, hence $\hat{\rho} - \hat{A} \leq 1$. These two inequalities imply that $tr(M) < 0$ for all η and k. According to the previous lemma, if the trace is negative, then the matrix can have an eigenvalue with a positive real part only

if its determinant is also negative, which occurs when $\eta\lambda_k^2 + \lambda_k(\eta\hat{A} - 3\hat{\rho}) < 0$. The problem thus reduces to determining the parameter values for which

$$\eta(\hat{A} + \lambda_k) - 3\hat{\rho} < 0. \tag{16}$$

The behaviour of the inequality can be illustrated on a plot, with $\hat{A}$ on the X-axis, and $\hat{\rho}$ on the Y-axis. The plots below show where the inequality is satisfied, for different values of the parameters η and k:



(a) $\eta = 1/2, k = 1$ (b) $\eta = 1/5, k = 1$ (c) $\eta = 1/5, k = 2$

Figure 4: The plots show, in red, the values of $(\hat{A}, \hat{\rho})$ for which inequality 16 is satisfied for fixed values of η and k. Subsequent plots illustrate how this region changes under the influence of changes in the parameters. The blue region indicates the region corresponding to realistic values of $\hat{A}$ and $\hat{\rho}$).

The red region shows where the inequality is satisfied, whereas the blue region corresponds to the realistic values of $\hat{A}, \hat{\rho}$. Outside the blue region, the system is unstable even without diffusion, so in order to observe a change in stability, we need to know where the blue and red regions overlap. It is easy to see that it occurs when $\eta < 3/\pi^2 k^2$. In particular, we can take $\hat{A} = 1, \hat{\rho} = 0.8$ and $\eta = 0.2$, which satisfy the inequality; the corresponding solution of the system is unbounded.

□

The unboundedness of the solution $(0, 0)$ of the linear system directly corresponds to the instability of the stationary solution $(\hat{A}, \hat{\rho})$ in the full model with diffusion. This is because the linearized system describes the behaviour of the full model under a small perturbation.

6 Numerical simulations of the model

6.1 On the interval $[0, l]$

In this chapter, the results of numerical simulations of the model on the interval $[0, l]$ are presented. The function χ is taken as the natural logarithm. This can approximately describe the behaviour of criminals in a small town where buildings are concentrated along a main road. In the simulations, the interval is divided into 100 elements, roughly corresponding to 100 houses. These simulations complement the earlier mathematical analyses of the model, allowing for a better understanding of its behaviour. The red dashed line in the plots represents the stationary solution values for the parameters used in each simulation. Unless stated otherwise, the simulations are carried out with parameters $\eta = 0.2, A^0 = \frac{1}{30}$ and $R = 1.6$, while the initial conditions are constant and equal to the corresponding stationary states, namely $A_0(x) \equiv A^0 + R \approx 1.63$, and $\rho_0(x) \approx \frac{R}{R+A^0} = 0.98$.

6.1.1 Model simulations for different initial conditions

The first simulation shows that when both A and ρ are initially set at their stationary values, they remain at these values throughout the simulation.

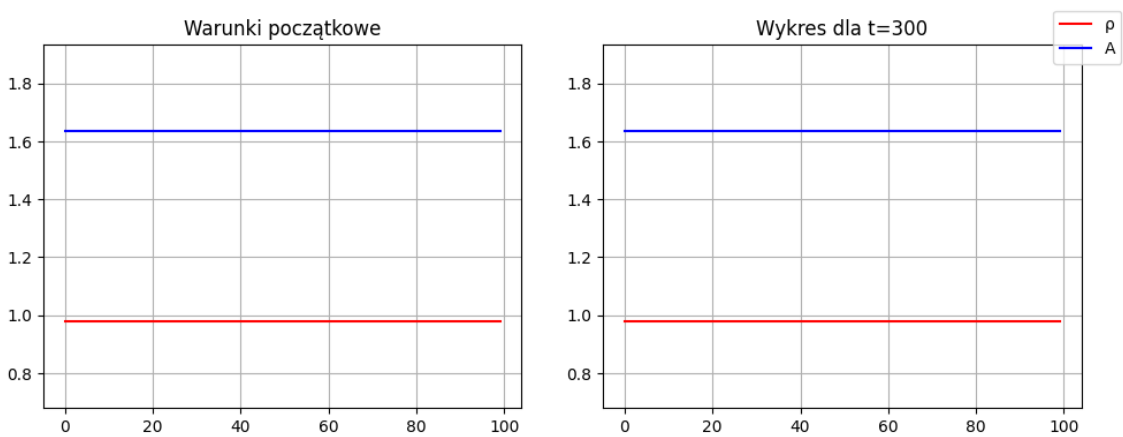
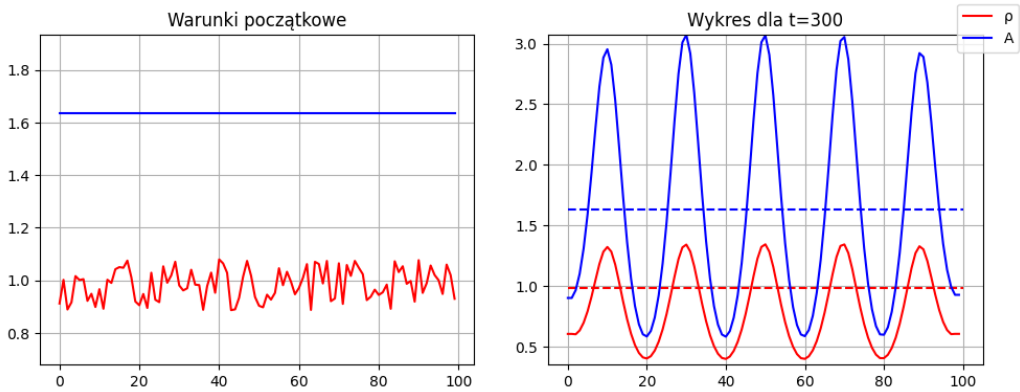


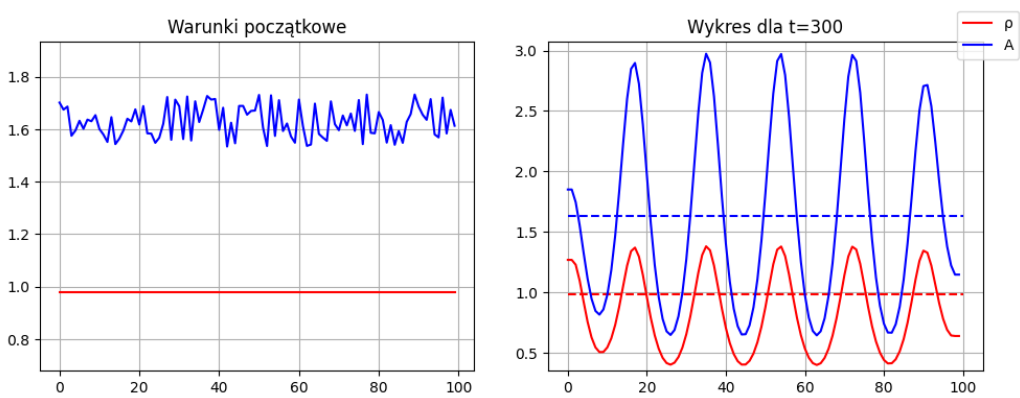
Figure 5: The simulation result for constant initial conditions, equal to the stationary solutions. This shows no change over time; the solution remains at the initial state.

However, when the initial condition for ρ is perturbed by white noise with variance 0.01 and mean 0, we observe oscillations around the stationary states. These oscillations are not symmetric in regard to the stationary states: the oscillations of ρ are

shifted downward relative to its stationary value, while those of A are slightly shifted upwards. A similar behaviour occurs when the initial condition for A is subjected to the same type of perturbation .



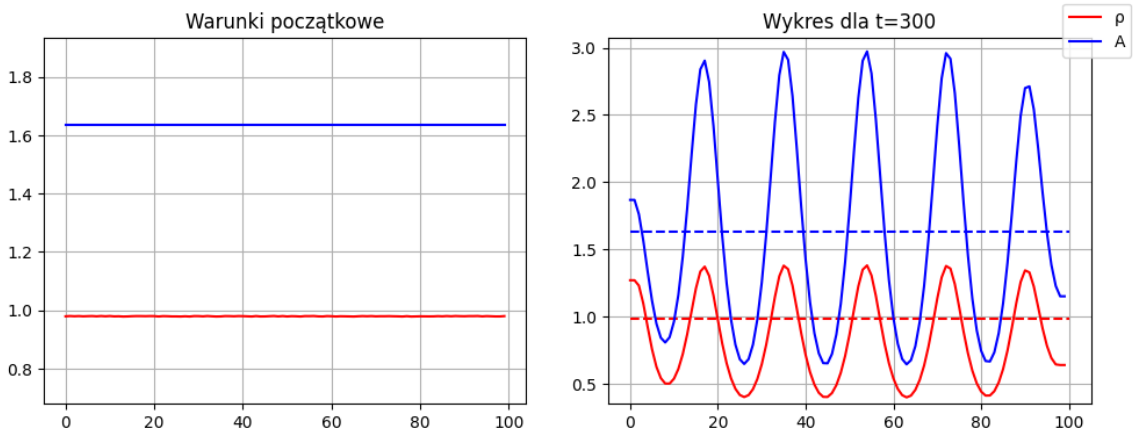
(a) Perturbed initial condition for ρ.



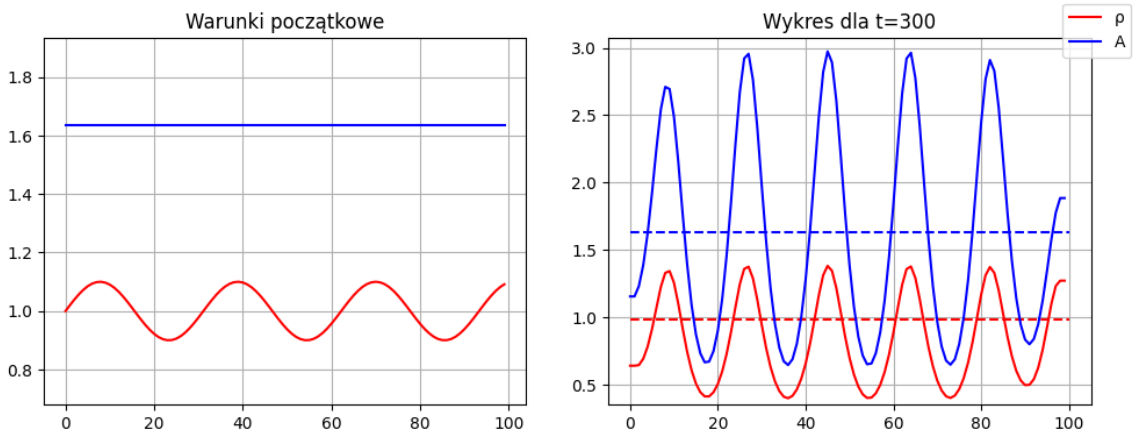
(b) Perturbed initial conditions for A.

Figure 6: Simulation results when the initial conditions is perturbed by white noise with variance 0.1.

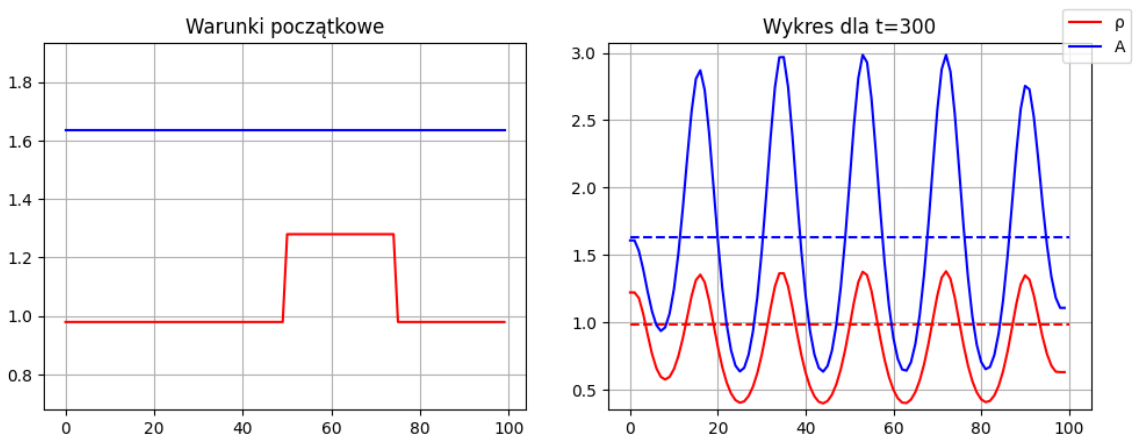
The simulation results are similar regardless of the type of perturbation. Below, this effect is illustrated for the initial condition ρ perturbed by white noise with variance 0.001, by a sine function with amplitude 0.2, and by a piecewise linear function. The results are shown only for perturbations of the initial state ρ, but, as before, analogous behaviour is observed for perturbations of A.



(a) Initial condition ρ perturbed by white noise with variance 0.001



(b) Initial condition ρ given by a sine function with amplitude 0.2, shifted by 1.



(c) Initial condition ρ shifted by 0.3 at 1/4 of the interval.

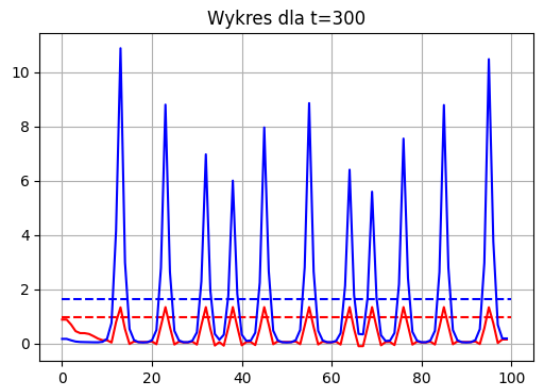
Figure 7: Simulation results of the model for various initial conditions of ρ.

The above simulations illustrate the occurence of Turing instability for the selected parameter values.

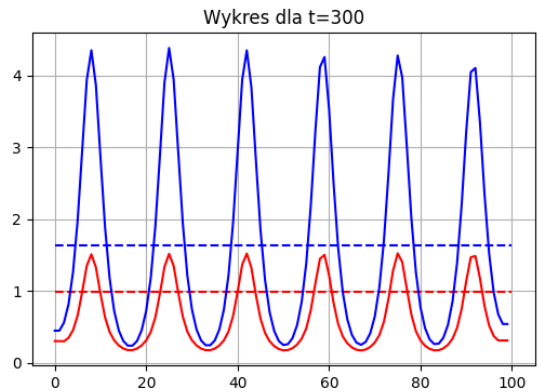
6.1.2 Parameter variations

The following simulations visualise how the solution changes with variations in the parameters η, A^0 and R. When the initial conditions are set at the equilibrium states, changes in the parameters have no effect. Therefore, the simulations presented below use the initial condition for ρ perturbed by white noise with variance 0.1, as shown previously in Figure (6a).

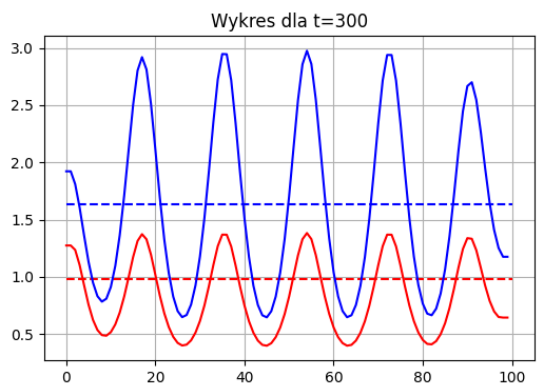
We first focus on analysing how the solutions change with variations in η. Recall that the parameters A^0 and R used in there simulations are $\frac{1}{30}$ and 1.6, respectively.



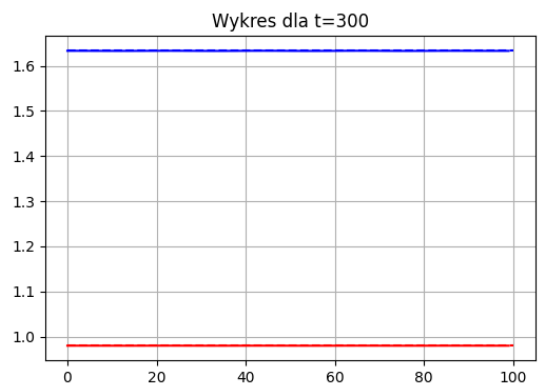
(a) $\eta = 0.01$



(b) $\eta = 0.1$



(c) $\eta = 0.02$

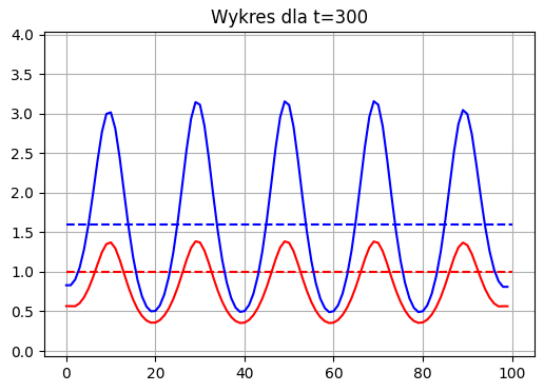


(d) $\eta = 0.03$

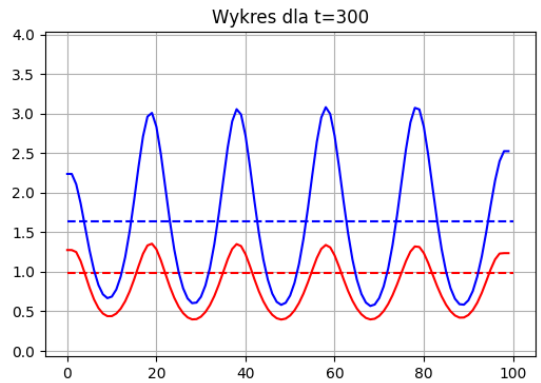
Figure 8: Simulation results of the model for different values of the parameter η, showing that the larger η is, the closer the solution remains to the stationary states.

The largest differences can be observed at two points: between Figures (a) and (b) - a change in the shape occurs around $\eta = 0.1$, which may be due to numerical instability; and between Figures (c) and (d) - a clear smoothing at the level of the stationary states, occurring between $\eta = 0.29$ and 0.3.

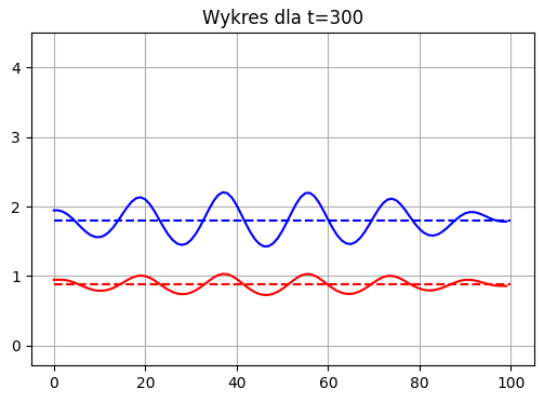
The next parameter whose influence we examine is A^0.



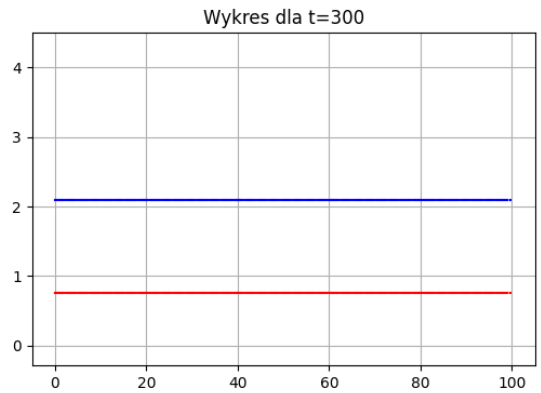
(a) $A^0 = 0$



(b) $A^0 = 1/30$



(c) $A^0 = 1/5$

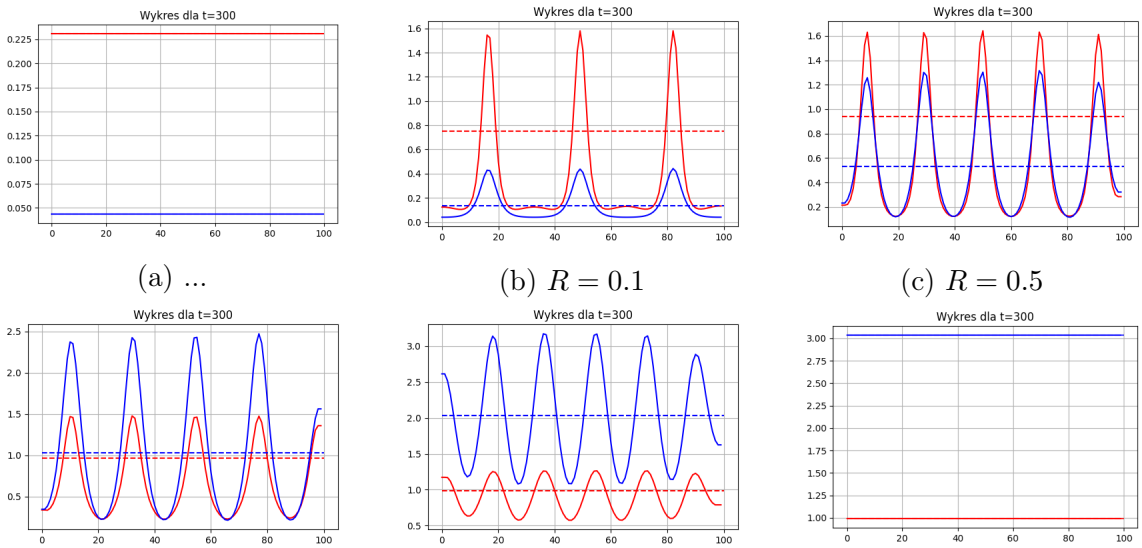


(d) $A^0 = 1/2$

Figure 9: Simulation results of the model for different values of the parameter A^0, showing that the larger A^0 is, the closer the solution remains to the stationary states.

We observe that increasing A^0 leads to smoother solutions. A sudden and pronounced smoothing at the level of the stationary states occurs around $A^0 = 0.22$. Naturally, changes in A^0 also affect the values of the stationary states.

The final parameter considered is R, which also influences the stationary states. The behaviour of the solution with respect to R is shown below.



(a) ... (b) $R = 0.1$ (c) $R = 0.5$

(d) $R = 1$ (e) $R = 2$ (f) $R = 3$

Figure 10: Simulation results of the model for different values of the parameter R.

In this case, a sudden smoothing also occurs at two points - when the parameter R drops below approximately 0.02 and when it exceeds approximately 2.41

6.2 On the square $[0, l] \times [0, l]$

In this part of the chapter, we present several visualisations of the behaviour of our model in $\mathbb{R}^2$. For these simulations, the initial condition primarily considered is a constant A_0 and ρ_0 perturbed by white noise with variance 0.1, as shown on Figure 11.

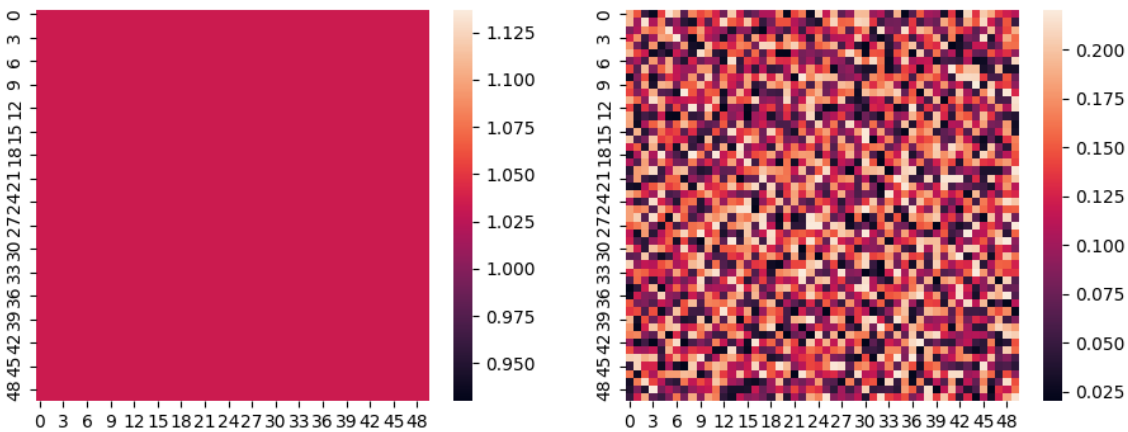
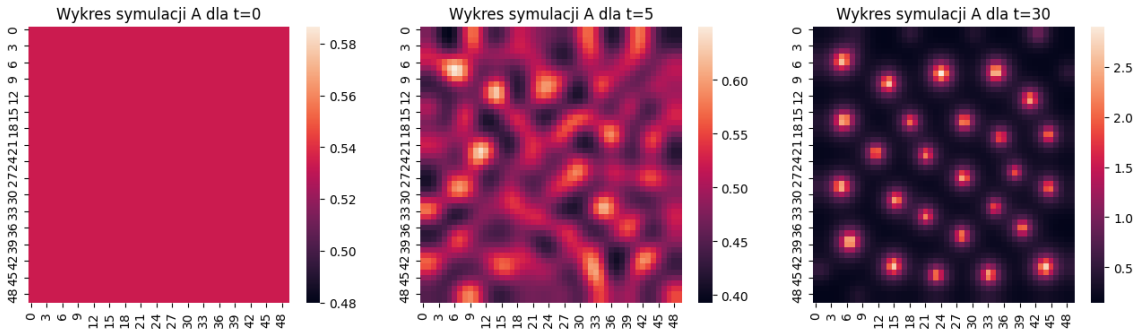
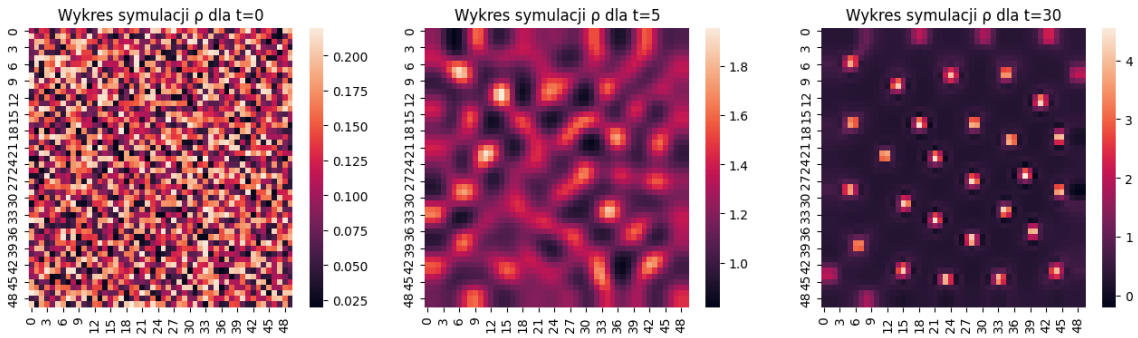


Figure 11: Initial conditions, for A (left) and ρ (right).

The baseline parameters were set to $\eta = 0.1, A^0 = 1/30, R = 0.5$. Unless a parameter change is specified for a particular simulation, it is assumed to take its baseline value. Below, the temporal evolution of the solutions for these baseline parameters is presented.



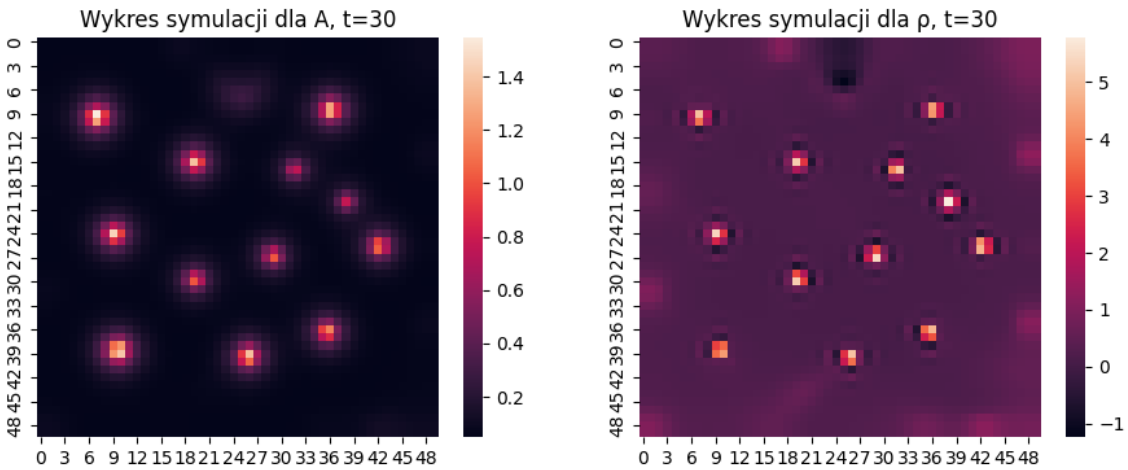
(a) Behaviour of A in time.



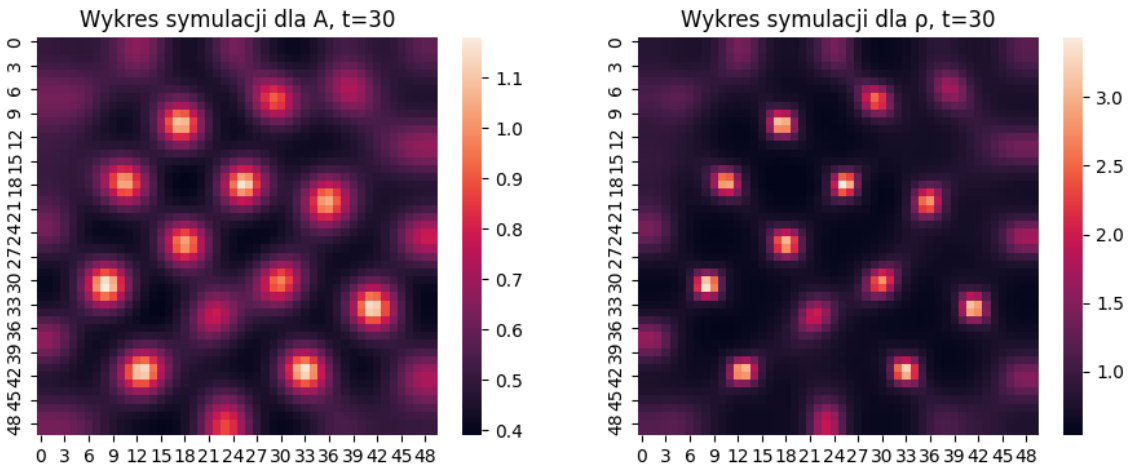
(b) Behaviour of ρ in time.

Figure 12: Subsequent stages of the model simulation for the baseline values $\eta = 0.1$, $A^0 = 1/30$ and $R = 0.5$.

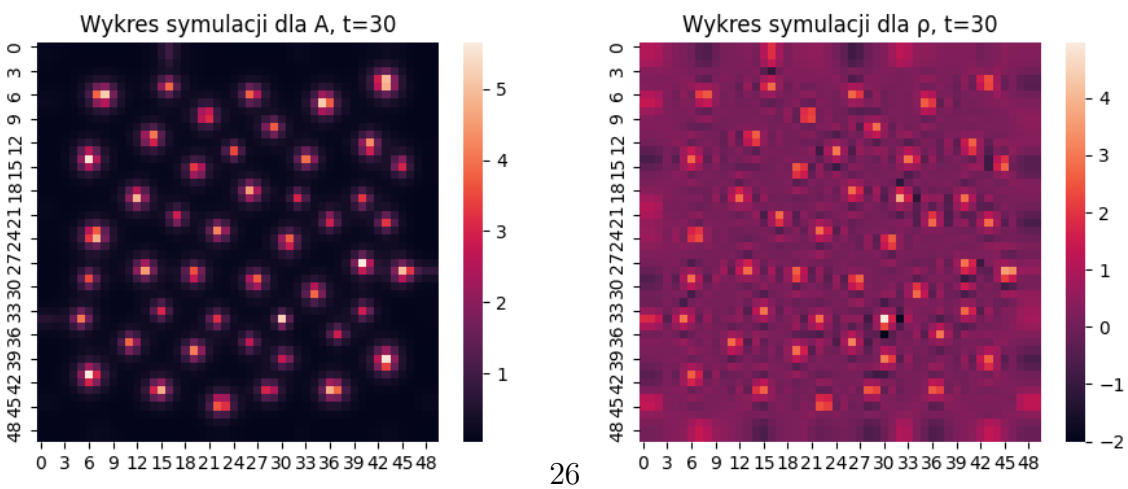
By varying the parameter values, we observe that larger values of R lead to denser concentrations in the plots, while their size changes proportionally with variations in η. Changes in parameter A^0 are not noticeably significant.



(a) $R = 0.1$



(b) $\eta = 0.2$



(c) $\eta = 0.03$

Figure 13: Simulation results for varying a single parameter: (a) decrease in R; (b) increase in η; (c) decrease in η.

Below, additional interesting simulation results are presented for a piecewise-constant initial condition for ρ, as well as for simultaneous changes in two parameters (using the previous initial condition).

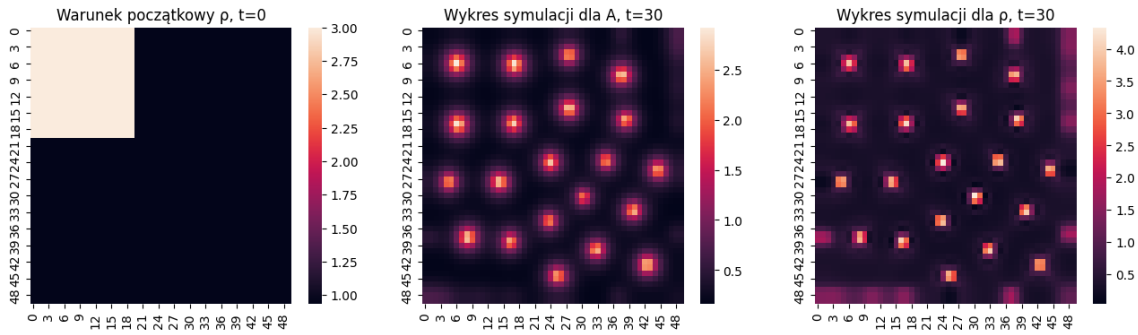
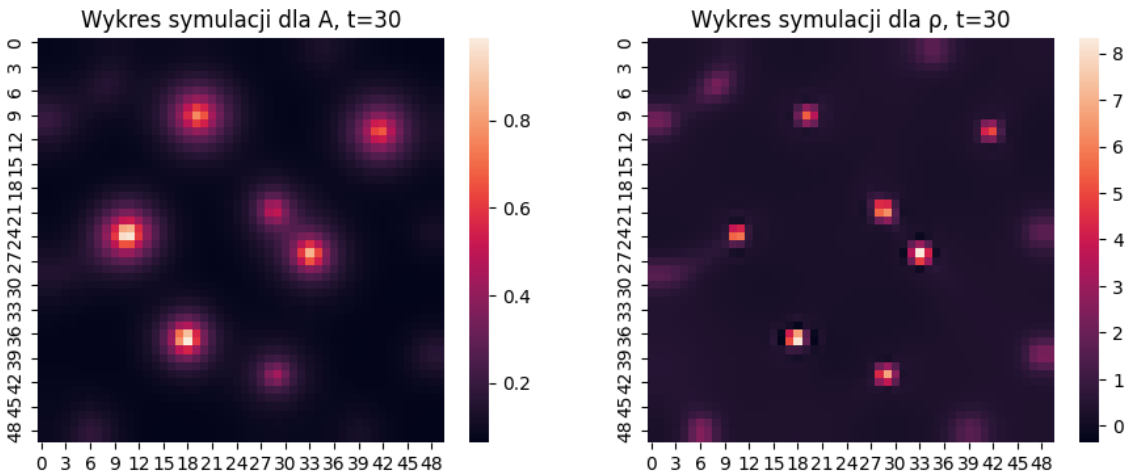
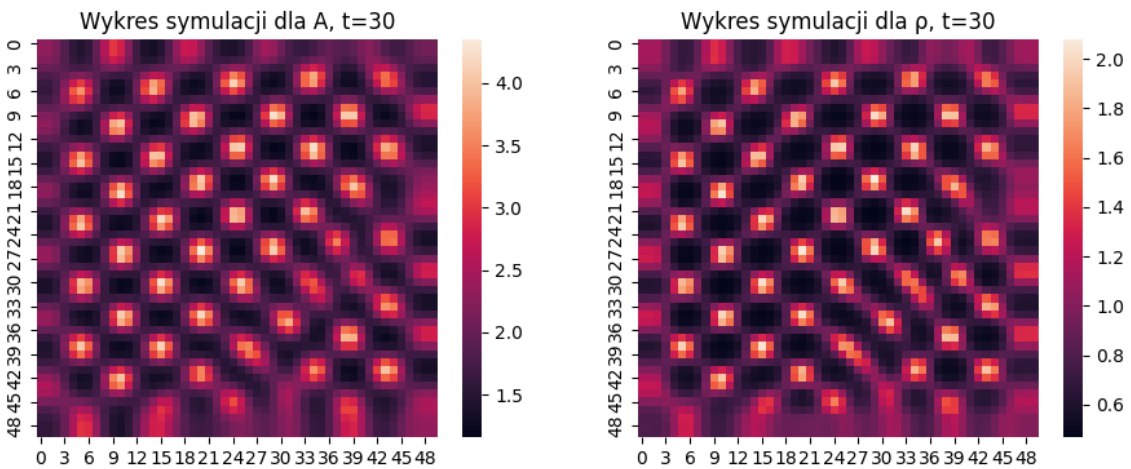


Figure 14: Boundary condition ρ_0 constant at 0 except in the upper-left corner, where the value is raised to 3.



(a) $\eta = 0.25, R = 0.1$



(b) $\eta = 0.05, R = 2$

Figure 15: Simulation results for varying parameters: (a) η increased and R decreased; (b) η decreased and R increased.

7 Programming Appendix

Below is the original code used to generate the numerical simulations presented in this work. The code was written in Python, primarily using the NumPy library:

```python
import pandas
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.animation as animation
import math
import tqdm
import seaborn as sns

def gradient(u, dx=1):
    res = u[2:]-u[:-2]
    return res/(2*dx)

def laplasjan(u, dx=1):
    dif = u[2:]+u[:-2]-2*u[1:-1]
    return dif/(dx**2)

dx = 1/5
dt = 1/100
l = 5*20
T = 300
eta = 0.27
A0 = 1/30
B = 1.6
A_0 = np.ones(l)*(A0+B)
ro_0 = np.ones(l)*ro0
eps = 0.1
ro_0 = np.maximum(eps*(2*np.random.random(l) - 1) + 0.1,0)

def symulacja(dx,dt,l,T,eta,A0,B,A_0,ro_0,t):
    time = int(T/dt)
    A = np.zeros((l,time))
    ro = np.zeros((l,time))
    A[:,0] = A_0
    ro[:,0] = ro_0
    for i in tqdm.tqdm(range(1, time)):
        # pierwsze rownanie z modelu
        A_next = eta * laplasjan(A[:,i-1],dx) - A[1:-1,i-1] + np.multiply(A[1:-1,i-1],ro[1:-1,i-1]) + A0
        A_next = A[1:-1,i-1] + dt*A_next
        A_next = np.append(A_next[0],np.append(A_next,A_next[-1]))
        A[:,i] = A_next
        # drugie rownanie z modelu
        A_log = np.log(np.maximum(A[:,i-1],0.01))
        ro_grad_A = np.multiply(ro[1:-1,i-1],gradient(A_log,dx))
        ro_grad_A = np.append(ro_grad_A[0],np.append(ro_grad_A,ro_grad_A[-1]))
        ro_next = 1*laplasjan(ro[:,i-1],dx) - 2*1*gradient(ro_grad_A,dx) + B - np.multiply(A[1:-1,i-1], ro[1:-1,i-1])
        ro_next = ro[1:-1,i-1] + dt*ro_next
        ro_next = np.append(ro_next[0],np.append(ro_next,ro_next[-1]))
        ro[:,i] = ro_next
    return A, ro

# 2D
dx = 1/5
dt = 1/100
l = 50
T = 30
eta = 0.1
A0 = 1/30
R = 0.16
A_0 = np.ones((l,l))*(A0+R)
eps = 0.1
ro_0 = np.maximum(eps*(2*np.random.random((l,l)) - 1) + 0.1,0)

def sym2d(dx,dt,l,T,eta,A0,R,A_0,ro_0):
    time = int(T/dt)
    A = np.zeros((l,l,time))
    ro = np.zeros((l,l,time))
    A[:,:,0] = A_0
    ro[:,:,0] = ro_0

    for i in tqdm.tqdm(range(1, time)):
        # pierwsze rownanie z modelu
        A_next_2d = eta*laplasjan_2d(A[:,:,i-1],dx)-A[1:-1,1:-1,i-1]+np.multiply(A[1:-1,1:-1,i-1],ro[1:-1,1:-1,i-1])+A0
        A_next_2d = A[1:-1,1:-1,i-1] + dt*A_next_2d
        A_next_2d = np.concatenate((A_next_2d[0,:].reshape(1,-1),A_next_2d,A_next_2d[-1,:].reshape(1,-1)))
```

```
        A_next_2d = np.hstack((A_next_2d[:,0].reshape(A_next_2d.shape[0],1),A_next_2d,A_next_2d[:,-1].reshape(A_next_2d.shape[0],1)))
        A[:,:,i] = A_next_2d
        # drugie rownanie z modelu
        grad_ro = gradient_2d(ro[:,:,i-1])
        grad_logA = gradient_2d(np.log(np.maximum(A[:,:,i-1],0.01)))
        ro_temp_x = 2*(grad_logA[0,:,:]*ro[1:-1,1:-1,i-1])
        ro_temp_y = 2*(grad_logA[1,:,:]*ro[1:-1,1:-1,i-1])
        gr_x = ro_temp_x[1:-1,2:]-ro_temp_x[1:-1,:-2]
        gr_y = ro_temp_y[2:,1:-1]-ro_temp_y[:-2,1:-1]
        gr_sum = gr_x+gr_y
        gr_sum = np.concatenate((gr_sum[0,:].reshape(1,-1),gr_sum,gr_sum[-1,:].reshape(1,-1)))
        gr_sum = np.hstack((gr_sum[:,0].reshape(gr_sum.shape[0],1),gr_sum,gr_sum[:,-1].reshape(gr_sum.shape[0],1)))
        ro_next_2d = laplasjan_2d(ro[:,:,i-1]) + R
        ro_next_2d = ro_next_2d - np.multiply(A[1:-1,1:-1,i-1],ro[1:-1,1:-1,i-1])
        ro_next_2d = ro_next_2d - 2*(gr_sum)
        ro_next_2d = ro[1:-1,1:-1,i-1] + dt*ro_next_2d
        ro_next_2d = np.concatenate((ro_next_2d[0,:].reshape(1,-1),ro_next_2d,ro_next_2d[-1,:].reshape(1,-1)))
        ro_next_2d = np.hstack((ro_next_2d[:,0].reshape(ro_next_2d.shape[0],1),ro_next_2d,ro_next_2d[:,-1].reshape(ro_next_2d.shape[0],1)))
        ro[:,:,i] = ro_next_2d

    return A, ro
```

References

[1] Chaohao Pan et al. „Crime modeling with truncated Lévy flights for residential burglary models". English. In: *Math. Models Methods Appl. Sci.* 28.9 (2018), pp. 1857–1880. ISSN: 0218-2025. DOI: 10.1142/S0218202518400080.

[2] Benoit Perthame. *Parabolic Equations in Biology. Growth, Reaction, Movement and Diffusion*. Springer, 2015. ISBN: 978-3-319-19499-8.

[3] M. B. Short, A. L. Bertozzi, and P. J. Brantingham. „Nonlinear patterns in urban crime: hotspots, bifurcations, and suppression". English. In: *SIAM J. Appl. Dyn. Syst.* 9.2 (2010), pp. 462–483. ISSN: 1536-0040. DOI: 10.1137/090759069.

[4] M. B. Short et al. „A statistical model of criminal behavior". English. In: *Math. Models Methods Appl. Sci.* 18 (2008), pp. 1249–1267. ISSN: 0218-2025. DOI: 10.1142/S0218202508003029.